A GLOBAL REGULARISATION OF THE GRAVITATIONAL *N*-BODY PROBLEM

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Abstract. The work of Aarseth and Zare (1974) is extended to provide a *global* regularisation of the classical gravitational three-body problem: by transformation of the variables in a way that does not depend on the particular configuration, we obtain equations of motion which are regular with respect to collisions between any pair of particles. The only cases excepted are those in which collisions between more than one pair occur simultaneously and those in which at least one of the masses vanishes. However, by means of the same principles the restricted problem is regularised globally if collisions between the two primaries are excluded. Results of numerical tests are summarised, and the theory is generalised to provide global regularisations, first, for perturbed three-body motion and, second, for the *N*-body problem. A way of increasing the number of degrees of freedom of a dynamical system is central to the method, and is the subject of an Appendix.

1. Introduction

In recent years, numerical experience with the classical gravitational *N*-body problem has motivated a vigorous search for methods of regularising the corresponding equations of motion. Serious numerical errors are associated with the r^{-2} singularity in the force between two particles separated by a distance r (Szebehely, 1968), and the purpose of regularisation is the transformation of singular equations into regular ones. It is also of advantage numerically if the regularised equations are stable, and theoretical developments are much facilitated if they are linear and if the regularisation is expressible in canonical form (Stiefel and Scheifele, 1971, pp. 1–2).

Since most numerical difficulties with the *N*-body problem are associated with close encounters between only two bodies, it should be possible to remove them by applying some regularisation of the perturbed two-body problem. The most successful is that of Kustaanheimo and Stiefel (KS-regularisation), whose application to the *N*-body problem is documented by Peters (1968), Aarseth (1971b), and Bettis and Szebehely (1971), but others have been used (Heggie, 1974).

In the sense of Szebehely (1967, p. 97), KS-regularisation is not 'global' when applied to the *N*-body problem with $N>2$, because the transformation to be used depends on which pair of particles is near collision. A global regularisation of the general three-body problem was provided some time ago by Lemaître (1952, 1955), although the Hamiltonian is complicated, and it is not clear that the equations for the motion of the plane containing the three particles really are regular. The sustained search for more suitable regularisations has also been encouraged by realisation of the dynamical and astrophysical importance of close approaches between more than two bodies (Aarseth, 1971a; Allen and Poveda, 1971; Heggie, 1972). Indeed, Waldvogel (1972) has produced a comparatively simple global regularisation for three-body

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motions confined to a plane, the Hamiltonian being a polynomial of degree 12 in the regularised coordinates and of degree two in the conjugate momenta.

Initiating a fresh approach, Aarseth and Zare (1974) recently obtained equations for the motion of two of the three bodies, say P_1 and P_2, relative to the third, P_3, with the property that they are regular for collisions both between P_1 and P_3 and between P_2 and P_3. However, a change of reference from P_3 to either P_1 or P_2 is required if equations regular with respect to collisions between P_1 and P_2 are sought; in other words, the regularisation is still not global.

In Section 2 of this paper we show how the method of Aarseth and Zare can be extended to yield a 'global' regularisation of the three-body problem in three dimensions; that is, by means of one short sequence of transformations, we obtain equations regular at points of collision between any pair of the three bodies. Following a discussion of numerical results in Section 3, Section 4 describes the modifications required in a treatment of the restricted problem. In Section 5 it is shown that the regularisation described in Section 2 can be applied to the motion of three bodies perturbed by $(N-3)$ others, where $N>3$, and in Section 6 a global regularisation of the N-body problem $(N\geqslant 2)$ is constructed.

The constant of gravitation is taken to be unity. Vectors will be column vectors, and a superscript T above a symbol for a vector or matrix will signify its transpose.

2. Global Regularisation of the Three-Body Problem

Let $\mathbf{q}_i'$ $(i=1, 2, 3)$ be the three 3-vectors of Cartesian coordinates of the three particles in some inertial frame. If $\mathbf{p}_i'$ $(i=1, 2, 3)$ are the conjugate momenta, the Hamiltonian H' is

$$H' = \sum_{i=1}^{3} \frac{1}{2m_i} \mathbf{p}_i'^2 - \frac{m_2 m_3}{|\mathbf{q}_2' - \mathbf{q}_3'|} - \frac{m_3 m_1}{|\mathbf{q}_3' - \mathbf{q}_1'|} - \frac{m_1 m_2}{|\mathbf{q}_1' - \mathbf{q}_2'|}, \tag{1}$$

where m_i $(i=1, 2, 3)$ are the three masses. We consider the solution of the corresponding Hamiltonian equations

$$\begin{aligned} \dot{\mathbf{q}}_i' &= \frac{1}{m_i} \mathbf{p}_i', \qquad (i = 1, 2, 3) \\ \dot{\mathbf{p}}_1' &= -\frac{m_1 m_2 (\mathbf{q}_1' - \mathbf{q}_2')}{|\mathbf{q}_1' - \mathbf{q}_2'|^3} - \frac{m_1 m_3 (\mathbf{q}_1' - \mathbf{q}_3')}{|\mathbf{q}_1' - \mathbf{q}_3'|^3}, \qquad (*) \end{aligned} \tag{2}$$

where a dot denotes differentiation with respect to time, t, and an asterisk beside an equation indicates that two similar equations are obtainable by cyclic interchange of indices. Let the initial conditions be

$$\begin{aligned} \mathbf{q}_i'(0) &= \mathbf{q}_{i0}', \qquad (i = 1, 2, 3) \\ \mathbf{p}_i'(0) &= \mathbf{p}_{i0}'. \qquad (i = 1, 2, 3) \end{aligned} \tag{3}$$

We now introduce variables $\mathbf{q}_i$ $(i=1, 2, 3)$ and $\langle\mathbf{q}\rangle$ defined by

$$\mathbf{q}_1 = \mathbf{q}_2' - \mathbf{q}_3', \qquad (*)$$

$$\langle\mathbf{q}\rangle = \frac{1}{M}\sum_{i=1}^{3} m_i\mathbf{q}_i', \tag{4}$$

where $M \equiv m_1 + m_2 + m_3$. If we followed the method of Aarseth and Zare, at this point we would transform the Hamiltonian to the variables $\mathbf{q}_1$, $\mathbf{q}_2$ and either $\mathbf{q}_3'$ or $\langle\mathbf{q}\rangle$, but clear progress towards a global regularisation would be made if $\mathbf{q}_3$ could be introduced on an equal footing with $\mathbf{q}_1$ and $\mathbf{q}_2$. The obvious difficulty, that $\mathbf{q}_1$, $\mathbf{q}_2$ and $\mathbf{q}_3$ are not independent vectors, must be removed somehow, and for the purposes of regularisation it is desirable to do so in the language of Hamiltonian dynamics. A different discussion of this problem, but in terms of Lagrangians, may be found in the work of Broucke and Lass (1973).

More precisely, our aim will be to introduce new variables $\mathbf{p}_i$ $(i=1, 2, 3)$ and $\langle\mathbf{p}\rangle$, which we will treat as conjugate momenta, and a new function $H(\mathbf{q}_i, \langle\mathbf{q}\rangle, \mathbf{p}_i, \langle\mathbf{p}\rangle)$, such that solution of Hamilton's equations with Hamiltonian H and suitable initial conditions allows us to recover the solution of (2) with initial conditions (3). In the next two paragraphs we shall show how to proceed in this direction with reasonable prospects for success, and only thereafter shall we prove formally that the new Hamiltonian system has the desired properties.

Because the variables $\mathbf{q}_i$ satisfy the condition $\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3 = \mathbf{0}$ they are not independent and our problem is not covered by the usual theory of point transformations (e.g. Goldstein, 1950, p. 244), but we can be guided by it. Accordingly, we consider new momenta satisfying

$$\mathbf{p}_i' = \sum_{j=1}^{3} \mathbf{p}_j^T \frac{\partial\mathbf{q}_j}{\partial\mathbf{q}_i'} + \langle\mathbf{p}\rangle^T \frac{\partial}{\partial\mathbf{q}_i'}\langle\mathbf{q}\rangle, \qquad (i = 1, 2, 3)$$

whence, by (4)

$$\mathbf{p}_1' = -\mathbf{p}_2 + \mathbf{p}_3 + \frac{m_1}{M}\langle\mathbf{p}\rangle. \qquad (*) \tag{5}$$

Substituting (4) and (5) into (1), and simplifying slightly, we take the new Hamiltonian to be

$$H = \frac{1}{2\mu_{23}}\mathbf{p}_1^2 + \frac{1}{2\mu_{31}}\mathbf{p}_2^2 + \frac{1}{2\mu_{12}}\mathbf{p}_3^2 + \frac{1}{2M}\langle\mathbf{p}\rangle^2 - \\ - \left(\frac{1}{m_1}\mathbf{p}_2^T\mathbf{p}_3 + \frac{1}{m_2}\mathbf{p}_3^T\mathbf{p}_1 + \frac{1}{m_3}\mathbf{p}_1^T\mathbf{p}_2\right) - \frac{m_2m_3}{q_1} - \frac{m_3m_1}{q_2} - \frac{m_1m_2}{q_3}, \tag{6}$$

where $q_i \equiv |\mathbf{q}_i|$ for $i=1, 2, 3$, and $\mu_{ij} \equiv m_im_j/(m_i+m_j)$.

Treating this as Hamiltonian, the corresponding Hamiltonian equations are

$$\dot{\mathbf{q}}_1 = \frac{1}{\mu_{23}}\mathbf{p}_1 - \frac{1}{m_2}\mathbf{p}_3 - \frac{1}{m_3}\mathbf{p}_2\,, \qquad (*)$$

$$\langle\dot{\mathbf{q}}\rangle = \frac{1}{M}\langle\mathbf{p}\rangle\,, \tag{7}$$

$$\dot{\mathbf{p}}_1 = -\frac{m_2 m_3}{q_1^3}\mathbf{q}_1\,, \qquad (*)$$

$$\langle\dot{\mathbf{p}}\rangle = \mathbf{0}\,.$$

For initial conditions we choose

$$\mathbf{q}_1(0) = \mathbf{q}'_{20} - \mathbf{q}'_{30}\,, \qquad (*)$$

$$\langle\mathbf{q}\rangle\,(0) = \frac{1}{M}\sum_{i=1}^{3} m_i q'_{i0}\,, \tag{8a}$$

following (3) and (4), and

$$\mathbf{p}_1\,(0) = \tfrac{1}{3}\left(\mathbf{p}'_{20} - \mathbf{p}'_{30} - \frac{m_2 - m_3}{M}\sum_{i=1}^{3}\mathbf{p}'_{i0}\right), \qquad (*)$$

$$\langle\mathbf{p}\rangle\,(0) = \sum_{i=1}^{3}\mathbf{p}'_{i0}\,, \tag{8b}$$

among the many possible solutions of (5) initially.

We have so far merely constructed a new dynamical problem whose relationship with the old one is not yet clear. However, we now show how the solution of the system (2) with initial conditions (3), may be recovered from a solution of the system (7) with (8) as initial conditions.

From (7), an integral of the motion is

$$\sum_{i=1}^{3}\mathbf{q}_i = \mathbf{0}\,, \tag{9}$$

using also (8a). This is the condition that (4) be invertible, for if we define

$$\mathbf{q}_1^* \equiv \frac{1}{M}(m_2\,\mathbf{q}_3 - m_3\mathbf{q}_2) + \langle\mathbf{q}\rangle\,, \qquad (*) \tag{10}$$

we then have

$$\mathbf{q}_1 = \mathbf{q}_2^* - \mathbf{q}_3^*\,. \qquad (*) \tag{11}$$

We also define, by analogy with (5),

$$\mathbf{p}_1^* \equiv -\,\mathbf{p}_2 + \mathbf{p}_3 + \frac{m_1}{M}\langle\mathbf{p}\rangle \qquad (*) \tag{12}$$

and now proceed to show that $\mathbf{q}_i^*$, $\mathbf{p}_i^*$ $(i=1, 2, 3)$ satisfy the Equations (2) and the boundary conditions (3), if $\mathbf{q}'_i$, $\mathbf{p}'_i$ are replaced, respectively, by starred variables.

From (7) and (10) we have

$$\dot{\mathbf{q}}_1^* = \frac{1}{m_1}\mathbf{p}_3 - \frac{1}{m_1}\mathbf{p}_2 + \frac{1}{M}\langle\mathbf{p}\rangle, \quad = \frac{1}{m_1}\mathbf{p}_1^* \qquad (*) \tag{13}$$

by (12), and we also find that

$$\dot{\mathbf{p}}_1^* = \frac{m_3 m_1}{q_2^3}\mathbf{q}_2 - \frac{m_1 m_2}{q_3^3}\mathbf{q}_3 \qquad (*)$$

by (7) and (12), whence

$$\dot{\mathbf{p}}_1^* = -\frac{m_1 m_3(\mathbf{q}_1^* - \mathbf{q}_3^*)}{|\mathbf{q}_1^* - \mathbf{q}_3^*|^3} - \frac{m_1 m_2(\mathbf{q}_1^* - \mathbf{q}_2^*)}{|\mathbf{q}_1^* - \mathbf{q}_2^*|^3} \qquad (*) \tag{14}$$

by (11). The initial conditions are, by (8a) and (10),

$$\mathbf{q}_i^*(0) = \mathbf{q}'_{i0} \qquad (i = 1, 2, 3) \tag{15}$$

and, with the aid of (8b) and (12),

$$\mathbf{p}_i^*(0) = \mathbf{p}'_{i0}\,. \qquad (i = 1, 2, 3) \tag{16}$$

Formulae (13)–(16) are to be compared with (2) and (3).

As presented here, this technique of enlarging a dynamical system may seem to succeed only fortuitously. It is in fact quite general, as will be shown in an Appendix. Its merit in this situation is that it has transformed the original Hamiltonian into one which looks like a Hamiltonian for three independent Keplerian systems, save for the coupling term represented in (6) by the crossed momentum products. The fact that the coupling term is linear in each momentum may well commend itself also in investigations other than ours, but for the present we continue towards our goal of a global regularisation of the three-body problem.

If we adopt the centre of mass frame, the term in $\langle\mathbf{p}\rangle$ in the new Hamiltonian can be dropped, and we suppose this to be done. We proceed to apply the regularising transformation of Kustaanheimo and Stiefel (Stiefel and Scheifele, 1971, p. 24) to each of the vectors $\mathbf{q}_i \equiv (q_{i1}, q_{i2}, q_{i3})$, obtaining the 4-vectors $\mathbf{Q}_i \equiv (Q_{i1}, Q_{i2}, Q_{i3}, Q_{i4})$, for $i = 1, 2, 3$. The treatment now follows closely that of Aarseth and Zare (1974) to which the reader is recommended to turn for more detail.

The relation between $\mathbf{q}_i$ and $\mathbf{Q}_i$ is given by

$$\begin{aligned} q_{i1} &= Q_{i1}^2 - Q_{i2}^2 - Q_{i3}^2 + Q_{i4}^2, \\ q_{i2} &= 2(Q_{i1}Q_{i2} - Q_{i3}Q_{i4}), \qquad (i = 1, 2, 3) \\ q_{i3} &= 2(Q_{i1}Q_{i3} + Q_{i2}Q_{i4}), \end{aligned} \tag{17}$$

and, if we introduce the matrix of partial derivatives, viz.

$$\mathbf{A}_i = 2\begin{pmatrix} Q_{i1} & Q_{i2} & Q_{i3} \\ -Q_{i2} & Q_{i1} & Q_{i4} \\ -Q_{i3} & -Q_{i4} & Q_{i1} \\ Q_{i4} & -Q_{i3} & Q_{i2} \end{pmatrix} \qquad (i = 1, 2, 3) \tag{18}$$

the new 4-momentum $\mathbf{P}_i$ $(i=1, 2, 3)$ is related to the old 3-momentum by the equation

$$\mathbf{p}_i = \frac{1}{4R_i}\mathbf{A}_i^T\mathbf{P}_i, \qquad (i=1, 2, 3) \tag{19}$$

where we set

$$R_i \equiv q_i \equiv \mathbf{Q}_i^T\,\mathbf{Q}_i. \qquad (i=1, 2, 3)$$

Note that the three quantities R_1, R_2 and R_3 are denoted in the paper of Aarseth and Zare by R_2, R_1 and R, respectively. We also have

$$\mathbf{p}_i^T\mathbf{p}_i = \frac{1}{4R_i}\mathbf{P}_i^T\mathbf{P}_i, \qquad (i=1, 2, 3)$$

and if these results be substituted into (6), the new Hamiltonian is obtained. We need not write it down, but we denote it by $\mathscr{H}(\mathbf{Q}_i, \mathbf{P}_i)$.

We now introduce a transformation of the independent variable by the differential equation

$$\mathrm{d}\tau = \frac{\mathrm{d}t}{R_1R_2R_3}, \tag{20}$$

τ being our new 'time'-variable. It is essentially the time-transformation used by both Lemaître and Waldvogel (*loc. cit.*), but remarks on the freedom of choice in the transformation will be found in the next section. The corresponding Hamiltonian is now

$$\Gamma^* \equiv (\mathscr{H} - h)\,R_1R_2R_3,$$

where h is the value of $\mathscr{H}$ along the particular solution path of interest. Recalling the definition of $\mathscr{H}$, we readily find that

$$\begin{aligned}\Gamma^* \equiv \frac{1}{8}\left(\frac{R_2R_3}{\mu_{23}}\mathbf{P}_1^T\mathbf{P}_1 + \frac{R_3R_1}{\mu_{31}}\mathbf{P}_2^T\mathbf{P}_2 + \frac{R_1R_2}{\mu_{12}}\mathbf{P}_3^T\mathbf{P}_3\right) - \\ - \frac{1}{16}\left(\frac{R_1}{m_1}\mathbf{P}_2^T\mathbf{A}_2\mathbf{A}_3^T\mathbf{P}_3 + \frac{R_2}{m_2}\mathbf{P}_3^T\mathbf{A}_3\mathbf{A}_1^T\mathbf{P}_1 + \frac{R_3}{m_3}\mathbf{P}_1^T\mathbf{A}_1\mathbf{A}_2^T\mathbf{P}_2\right) - \\ - m_2m_3R_2R_3 - m_3m_1R_3R_1 - m_1m_2R_1R_2 - hR_1R_2R_3.\end{aligned}$$

This is the final form of the Hamiltonian: a polynomial of degree six in the regularised coordinates and degree two in the conjugate momenta.

Hamilton's equations become

$$\mathbf{Q}_i' = \frac{\partial\Gamma^*}{\partial\mathbf{P}_i}, \qquad \mathbf{P}_i' = -\frac{\partial\Gamma^*}{\partial\mathbf{Q}_i}, \qquad (i=1, 2, 3) \tag{21}$$

where a prime now denotes differentiation with respect to τ. We see that these equations are regular with respect to collisions between any pair of bodies, as required. Furthermore, this has been achieved by one transformation, viz. the composition of $\mathbf{q}_i' \to \mathbf{q}_i$ and $\mathbf{q}_i \to \mathbf{Q}_i$, together with $t \to \tau$, and so the regularisation is global. It is also expressed in Hamiltonian form, and therefore satisfies all the desiderata set out in the

introduction except, possibly, stability and linearity. In fact, the equations for $\mathbf{P}_i'$ and $\mathbf{Q}_i'$ are linear in $\mathbf{P}_i$ and $\mathbf{Q}_i$ for each $i=1, 2, 3$.

3. Numerical Experience

A computer programme has been written for the purpose of integrating numerically Equations (20) and (21), and performing the transformations of the coordinates and momenta whenever these are required. The method has been tested on certain interesting cases against the three-body regularisation of Aarseth and Zare (1974), both programmes using the Runge-Kutta-Fehlberg 7(8) integration algorithm (Fehlberg, 1968). The same computer, in which numbers are stored to about sixteen significant figures, was used for both programmes. It should be noted that the computing time for one step in the integration is longer by a factor of about 1.6 with the present regularisation, because there are more equations to integrate and the right-hand sides of (21) are more complicated than the corresponding expressions in the method of Aarseth and Zare.

First, we consider the 'Pythagorean problem' solved by Szebehely and Peters (1967), and present results for integrations up to time $t=16$. In Table I are given the number of integration steps used and the corresponding relative error in the energy, i.e. $|\delta E/E|$,

TABLE I

Pythagorean problem

Tolerance	Equations (21)		Aarseth and Zare	
	Number of steps	$\lvert\delta E/E\rvert$	Number of steps	$\lvert\delta E/E\rvert$
10^{-8}	147	2×10^{-10}	129	2×10^{-11}
10^{-10}	257	7×10^{-13}	225	2×10^{-12}
10^{-12}	455	2×10^{-13}	397	1×10^{-13}

at time $t=16$. Different numbers of steps are obtained by imposing different tolerances on the truncation error at each step, as estimated in the integration algorithm itself, and the values used are given in the first column of the table. Limitations due to rounding errors are beginning to impose themselves at the largest step number. It should also be pointed out that, in each case, the energy error at $t=16$ was not less than either of those at $t=4$, 8 or 12, except for the 129-step integration of the Aarseth-Zare method, where it reached 7×10^{-11} at $t=4$. Accordingly, it would seem that comparable accuracy is achieved with both methods for approximately equal numbers of steps, and so, integration steps taking longer with the present method, it is less efficient for this problem.

The next case for comparison is Example II of Aarseth and Zare (1974). As the initial conditions lead to an encounter between a binary and a third body, it is a realis-

TABLE II

'Realistic' problem

Tolerance	Equations (21)		Aarseth and Zare	
	Number of steps	$\lvert\delta E/E\rvert$	Number of steps	$\lvert\delta E/E\rvert$
10^{-8}	48	3×10^{-9}	42	4×10^{-9}
10^{-10}	83	3×10^{-11}	70	3×10^{-11}
10^{-12}	149	3×10^{-13}	124	2×10^{-13}

tic problem in the context of stellar dynamics. In Table II $|\delta E/E|$ is given after the lapse of 0.002 time units. The present method emerges from this comparison also as significantly the weaker of the two.

In addition to (20), the time-transformation

$$\mathrm{d}\tau = \frac{(R_1+R_2+R_3)^n}{R_1R_2R_3}\,\mathrm{d}t, \tag{22}$$

with $n=\frac{3}{2}$, has been investigated. Although the polynomial form of the regularised Hamiltonian is now ruined, the result is still a global regularisation. Hamilton's equations become

$$\mathbf{Q}_i' = (R_1+R_2+R_3)^{-3/2}\frac{\partial\Gamma^*}{\partial\mathbf{P}_i} \qquad (i=1,2,3) \tag{23}$$

and

$$\mathbf{P}_i' = -(R_1+R_2+R_3)^{-3/2}\left\{\frac{\partial\Gamma^*}{\partial\mathbf{Q}_i} - \tfrac{3}{2}\Gamma^*\frac{\partial}{\partial\mathbf{Q}_i}\ln(R_1+R_2+R_3)\right\} \qquad (i=1,2,3) \tag{24}$$

but, because $\Gamma^*=0$ on the solution path, the last equation may be replaced by

$$\mathbf{P}_i' = -(R_1+R_2+R_3)^{-3/2}\frac{\partial\Gamma^*}{\partial\mathbf{Q}_i} \qquad (i=1,2,3) \tag{25}$$

and the equations of motion are scarcely more complicated than before.

With the 'realistic' problem discussed above and a tolerance of 10^{-10}, use of Equations (22), (23) and (25) reduced the relative error in energy to 1×10^{-11} and the number of steps to 63. Some improvements have been noted by Dr Aarseth when an analogous modification is made to the Aarseth-Zare method. For the Pythagorean problem also, Equations (22), (23) and (25) yielded somewhat improved results in general.

Although there is no reason to suppose that a method designed to regularise two-body collisions will improve the integration of motions close to triple collisions, it is of interest to try. For a triple collision at time $t=0$, we have, asymptotically as $t\to0-$, the relations

$$R_i \sim \alpha_i t^{2/3} \qquad (i=1,2,3)$$

for certain constants α_i (Siegel and Moser, 1971, § 12) and if τ is defined by (22) we

deduce that it is bounded as $t \to 0-$ if and only if $n > \frac{3}{2}$. On the other hand, a necessary and sufficient condition that the equations of motion be regular as $t \to 0-$ is $n < \frac{3}{2}$, and so a compromise was sought by the choice $n = \frac{3}{2}$.

In practice, no matter whether Equations (20) and (21) or Equations (22), (23) and (25) are used, numerical integrations are very unsatisfactory if a 'very close' triple encounter occurs, in the sense that the mutual distances of the three bodies are simultaneously less than about 10^{-7} of their initial values. The character of the solutions is then dynamically implausible and changes drastically when the tolerance is improved; furthermore, they compare badly both in these respects and in detail with the results produced by the method of Aarseth and Zare.

This difficulty is immediately removed if we replace Equation (25) by (24) and integrate Equations (22)–(24) for, although the two equations are formally identical on the solution path, their numerical behaviour and stability are quite different. This fact can be understood by solving the linearised equations for infinitesimal coplanar deviations from one of the exact analytic three-body collision orbits, in much the same manner as that described for unregularised equations by Siegel and Moser (1971, §13). The details of the calculation are quite tedious and will not be dwelt upon, but the results are informative. It is found that the introduction of extra degrees of freedom, in the transition from (1) to (6), leads to new modes, but none of these are unstable as triple collision is approached. It is the time-transformation which can destabilise certain modes: for Equations (20) and (21), and for the set (22), (23) and (25), modes are present which grow as R_i^{-1}, where R_i is any one of the mutual distances. No such unstable mode is present if Equations (22)–(24) are adopted.

For practical purposes, the stabilising effect of the extra term removed in the transition from (24) to (25) may be likened to that of the control term introduced by Baumgarte (1972) into the equations for two-body motion. There too it arises from a canonical time-transformation.

With Equations (22)–(24) and a tolerance of 10^{-12}, the three-body systems used by Aarseth and Zare to study close triple encounters have been integrated. All masses are unity, and the bodies are initially at rest at the points having coordinates $(-1, 0, 0)$, $(1, 0, 0)$, $(\Delta x, \sqrt{3}+\Delta y, 0)$ with respect to rectangular axes, where $\sqrt{3}$ was entered rounded to 15 decimal places, and Δx and Δy are given in the first two columns of Table III. After the lapse of the number of integration steps given in the third column, the signs of the momentum components were reversed and the integration continued for the same length of physical time t as that which had elapsed during the first section of the integration. The accuracy with which the initial conditions were recovered is indicated by the figure in the last column, $(\delta x_i, \delta y_i, 0)$ being the error in the coordinate vector for the ith body $(i = 1, 2, 3)$, and the fourth column shows one third of the minimum value, C, during the whole integration, of the perimeter of the triangle defined by the three bodies. In the first five cases, a binary had formed by the time at which the momenta were reversed, and column five shows the ratio of the separation from it of the receding third body to the semi-major axis. With the last case, reversal of the momenta was chosen to occur at the step following that at which the minimum

TABLE III

Close triple encounters

Δx	Δy	N_{steps}	$C/3$	R_f/a_f	$\{\frac{1}{3}\sum_{i=1}^{3}(\delta x_i^2+\delta y_i^2)\}^{1/2}$
0	10^{-4}	250	4×10^{-4}	32	3×10^{-11}
0	10^{-6}	250	7×10^{-6}	30	5×10^{-10}
0	10^{-8}	250	1×10^{-7}	28	3×10^{-9}
10^{-4}	0	250	3×10^{-4}	31	6×10^{-11}
10^{-6}	0	250	6×10^{-6}	29	7×10^{-10}
0	0	139	2×10^{-13}	–	4×10^{-6}

value, C, was reached. If the integration was continued much beyond this point, a binary was formed in this case also, but by then the bilateral symmetry of the configuration had been lost because of integration errors.

A comparison with the corresponding table of Aarseth and Zare (1974) clearly reveals the advantage secured by use of the present method in the first five cases. For the same number of steps, the integration proceeds further with the present regularisation, as judged by the values of R_f/a_f, and yet the results of the time-reversal are better. In the last case, fewer steps are taken than in the integration of Aarseth and Zare, and yet the three particles here approach more closely as measured by C. Accordingly, the orbit here is a more difficult one to integrate and the time-reversal is less satisfactory.

In an assessment of this information, it should be observed that each integration step with the system (22)–(24) takes slightly longer, by about 15%, than with Equations (20) and (21). However, it is noteworthy that the success of the time-reversals recorded in Table III does not depend on any judicious choice for the initial labelling of the bodies, which is the case with the method of Aarseth and Zare.

On the basis of the evidence from the Pythagorean and 'realistic' problems, it is clear that Equations (20) and (21) are sufficiently powerful for most purposes, but in situations where very close triple encounters are anticipated it becomes advisable to adopt (22)–(24). Indeed, limited experience suggests that this last set of equations is the best of those considered in this paper even for less critical cases such as the Pythagorean problem.

4. The Restricted Three-Body Problem

Because of the presence in the regularised Hamiltonian Γ^* of terms containing the reciprocals of the masses, the Equations (21) are inapplicable in case any of the masses vanishes, and, in particular, to the restricted three-body problem. So far it has not proved possible to rewrite them in a form which is at once suitable for the restricted problem and regular with respect to all possible binary collisions between the three bodies, but the results of Section 2 may be recast to treat the restricted problem if we are content to forego regularity during collisions between the two primaries. Indeed,

all that is usually meant by a 'global' regularisation in the context of this problem (Szebehely, 1967, p. 97) is a process yielding a set of equations regular with respect to collisions between the massless particle and either primary.

We consider the limit $m_3 \to 0$, and begin by introducing new 4-vectors $\mathbf{V}_i$ by the definitions

$$\mathbf{V}_i = \frac{1}{m_3}\mathbf{P}_i. \qquad (i = 1, 2) \tag{26}$$

The first of Equations (21) become

$$\mathbf{Q}_1' = \frac{1}{4}\frac{R_2R_3(m_2+m_3)}{m_2}\mathbf{V}_1 - \tfrac{1}{16}\mathbf{A}_1\left(\frac{R_2}{m_2}\mathbf{A}_3^T\mathbf{P}_3 + R_3\mathbf{A}_2^T\mathbf{V}_2\right) \quad (\dagger) \tag{27}$$

and

$$\mathbf{Q}_3' = \frac{1}{4}\frac{R_1R_2}{\mu_{12}}\mathbf{P}_3 - \tfrac{1}{16}m_3\mathbf{A}_3\left(\frac{R_1}{m_1}\mathbf{A}_2^T\mathbf{V}_2 + \frac{R_2}{m_2}\mathbf{A}_1^T\mathbf{V}_1\right), \tag{28}$$

the dagger (†) after (27) indicating that a similar equation is obtained by interchanging the suffices 1 and 2. These equations are well-behaved in the limit $m_3 \to 0$.

The equations for the momentum are a trifle more awkward and introduce a new feature. We define by the identity

$$m_3k \equiv h - \frac{1}{2\mu_{12}}\mathbf{p}_3^2 + \frac{m_1m_2}{q_3} \tag{29}$$

a new variable k, whose physical meaning will be clearer when an alternative derivation is described later. Using (7), we observe that

$$m_3\dot{k} = \frac{1}{q_3^3}\mathbf{q}_3^T(m_2\mathbf{p}_2 + m_1\mathbf{p}_1) \tag{30}$$

and since, by (17) and (18),

$$\mathbf{q}_i \equiv \tfrac{1}{2}\mathbf{A}_i^T\mathbf{Q}_i, \qquad (i = 1, 2, 3)$$

we may use (19), (20), (26) and (30) to deduce that

$$k' = \frac{1}{8R_3^2}\mathbf{Q}_3^T\mathbf{A}_3(m_2R_1\mathbf{A}_2^T\mathbf{V}_2 + m_1R_2\mathbf{A}_1^T\mathbf{V}_1), \tag{31}$$

which is independent of m_3 but singular as $R_3 \to 0$, i.e. in case of collision between the primaries.

It is easy to deduce from the second of Equations (21), and (26), that

$$\mathbf{V}_1' = 2\mathbf{Q}_1\left(kR_3R_2 - \frac{1}{8}\frac{R_3(m_1+m_3)}{m_1}\mathbf{V}_2^2 + \frac{1}{16m_1}\mathbf{V}_2^T\mathbf{A}_2\mathbf{A}_3^T\mathbf{P}_3 + m_1R_3\right) + \\ + \frac{1}{16}\left(\frac{R_2}{m_2}\mathbf{P}_3^T\mathbf{A}_3 + R_3\mathbf{V}_2^T\mathbf{A}_2\right)\frac{\partial\mathbf{A}_1^T}{\partial\mathbf{Q}_1}\mathbf{V}_1 \qquad (\dagger) \tag{32}$$

and

$$\mathbf{P}_3' = 2\mathbf{Q}_3\left(hR_1R_2 - \frac{1}{8}\frac{R_2m_3(m_2+m_3)}{m_2}\mathbf{V}_1^2 - \frac{1}{8}\frac{R_1m_3(m_1+m_3)}{m_1}\times\right.$$
$$\left.\times\,\mathbf{V}_2^2 + \tfrac{1}{16}m_3\mathbf{V}_1^T\mathbf{A}_1\mathbf{A}_2^T\mathbf{V}_2 + m_1m_3R_1 + m_2m_3R_2\right) +$$
$$+\,\tfrac{1}{16}m_3\left(\frac{R_1}{m_1}\,\mathbf{V}_2^T\mathbf{A}_2 + \frac{R_2}{m_2}\,\mathbf{V}_1^T\mathbf{A}_1\right)\frac{\partial\mathbf{A}_3^T}{\partial\mathbf{Q}_3}\,\mathbf{P}_3\,. \tag{33}$$

Again these equations are well-behaved as $m_3 \to 0$, and indeed the last simplifies enormously if $m_3 = 0$.

We have yet to specify initial conditions in a way suitable for small or vanishing values of m_3. There is no difficulty with the coordinates, but in place of certain of the momenta it is preferable to adopt new variables $\mathbf{v}_3'$ and $\mathbf{v}_i$ $(i=1, 2)$, defined by the expressions

$$\begin{aligned}\mathbf{p}_3' &\equiv m_3\mathbf{v}_3'\,,\\ \mathbf{p}_i &\equiv m_3\mathbf{v}_i\,. \qquad (i=1,2)\end{aligned} \tag{34}$$

The quantities $\mathbf{v}_i$ are related to $\mathbf{V}_i$ $(i=1, 2)$ by the equations

$$\mathbf{v}_i = \frac{1}{4R_i}\mathbf{A}_i^T\mathbf{V}_i\,, \qquad (i=1,2)$$

obtained from (19), (26) and (34).

Equations (5) now become

$$\mathbf{p}_1' = -\,m_3\mathbf{v}_2 + \mathbf{p}_3 + \frac{m_1}{M}\langle\mathbf{p}\rangle,$$

$$\mathbf{p}_2' = -\,\mathbf{p}_3 + m_3\mathbf{v}_1 + \frac{m_2}{M}\langle\mathbf{p}\rangle,$$

$$\mathbf{v}_3' = -\,\mathbf{v}_1 + \mathbf{v}_2 + \frac{1}{M}\langle\mathbf{p}\rangle,$$

and these must be satisfied given initial values $\mathbf{p}_{10}'$, $\mathbf{p}_{20}'$ and $\mathbf{v}_{30}'$ for the variables on the left. The solution given by (8b) is not suitable for $m_3 \to 0$, but it will be recalled that we had considerable freedom of choice. Let us adopt, therefore, the initial values

$$\mathbf{v}_{10} = \frac{1}{M}(\mathbf{p}_{20}' - m_2\mathbf{v}_{30}'),$$

$$\mathbf{v}_{20} = \frac{1}{M}(-\,\mathbf{p}_{10}' + m_1\mathbf{v}_{30}'),$$

$$\mathbf{p}_{30} = \frac{1}{M}(m_2\mathbf{p}_{10}' - m_1\mathbf{p}_{20}'),$$

$$\langle\mathbf{p}\rangle_0 = \mathbf{p}_{10}' + \mathbf{p}_{20}' + m_3\mathbf{v}_{30}'\,,$$

except that we select a frame in which the last expression vanishes.

To obtain the initial value of k when m_3 is very small, it is advisable to rewrite (29). Replacing h by the expression on the right-hand side of (6), and using (34), it is easy to

see that

$$k = \frac{1}{2}\left(\frac{m_2 + m_3}{m_2}\,\mathbf{v}_1^2 + \frac{m_1 + m_3}{m_1}\,\mathbf{v}_2^2\right) - \left(\frac{1}{m_1}\,\mathbf{v}_2^T\mathbf{p}_3 + \frac{1}{m_2}\,\mathbf{v}_1^T\mathbf{p}_3 + \mathbf{v}_1^T\mathbf{v}_2\right) - \\ - \frac{m_2}{q_1} - \frac{m_1}{q_2}$$

in the rest-frame of the centre of mass. This equation is suitable for computation of the initial value of k.

The derivation of the results given in this section was motivated by the search for a global regularisation of the restricted problem, which is the case $m_3 = 0$. Nevertheless, the equations have been given in full for the case $m_3 \neq 0$ because it is likely that they represent a more suitable form of the regularisation than that presented in Section 2 in the case when one of the masses is very small but not necessarily zero. From this statement we must exclude the case of orbits in which the two massive bodies collide. Regarded as a regularisation of the restricted problem, the equations obtained here may be used to treat three-dimensional motion of the massless particle, and the relative motion of the primaries may be hyperbolic, parabolic or elliptic, as well as circular; collision orbits alone being excepted.

We now give an alternative derivation of the regularisation for the case of the restricted problem, for it reveals the meaning of the quantity k, it illustrates an interesting extension of the technique for extending the number of degrees of freedom of a dynamical system, and it can be generalised readily to the case of a massless particle moving under the attraction of any number of other bodies. We begin with the Hamiltonian per unit mass for the motion of the massless third body, viz.

$$H' = \tfrac{1}{2}\mathbf{p}'^2 - \frac{m_1}{|\mathbf{q}' - \mathbf{q}_1'|} - \frac{m_2}{|\mathbf{q}' - \mathbf{q}_2'|}, \tag{35}$$

where $\mathbf{q}_i'$ $(i=1, 2)$ are the Cartesian position vectors of the primaries, given as functions of time, m_i $(i=1, 2)$ are their masses, $\mathbf{q}'$ is the position vector of the third body, and $\mathbf{p}'$ is the conjugate momentum.

As in Section 2 we shall try to find a Hamiltonian and suitable momentum variables such that the variables $\mathbf{q}_i$, defined by

$$\mathbf{q}_i \equiv \mathbf{q}_i' - \mathbf{q}_i', \qquad (i = 1, 2) \tag{36}$$

may be treated as independent. We proceed, as before, along the course we would follow if performing a normal point transformation, and write down a generating function, W, where

$$W \equiv (\mathbf{q}' - \mathbf{q}_1')^T\,\mathbf{p}_1 + (\mathbf{q}' - \mathbf{q}_2')^T\,\mathbf{p}_2\,. \tag{37}$$

Here, $\mathbf{p}_i$ will be the variables conjugate to $\mathbf{q}_i$ $(i=1, 2)$. We then find that we should take

$$\mathbf{p}' = \frac{\partial W}{\partial \mathbf{q}'} = \mathbf{p}_1 + \mathbf{p}_2 \tag{38}$$

and the new Hamiltonian will be

$$H = H' + \frac{\partial W}{\partial t},$$

$$= \tfrac{1}{2}(\mathbf{p}_1^2 + \mathbf{p}_2^2) + \mathbf{p}_1^T\mathbf{p}_2 - \frac{m_1}{q_1} - \frac{m_2}{q_2} - \dot{\mathbf{q}}_1'^T\mathbf{p}_1 - \dot{\mathbf{q}}_2'^T\mathbf{p}_2, \tag{39}$$

by (35)–(38). Here is a new feature, for we have introduced coordinate transformations (36) dependent explicitly on the time, and the Hamiltonian is altered in consequence.

If the initial values of the old variables are $\mathbf{q}_0'$, $\mathbf{q}_{i0}'$ $(i=1, 2)$ and $\mathbf{p}_0'$, (36) obliges us to choose initially

$$\mathbf{q}_i(0) = \mathbf{q}_0' - \mathbf{q}_{i0}', \qquad (i = 1, 2) \tag{40}$$

and (38) may be satisfied initially with

$$\mathbf{p}_i(0) = \tfrac{1}{2}\mathbf{p}_0'. \qquad (i = 1, 2) \tag{41}$$

The equations of motion for the Hamiltonian (39) are

$$\dot{\mathbf{q}}_i = \mathbf{p}_1 + \mathbf{p}_2 - \dot{\mathbf{q}}_i' \qquad (i = 1, 2)$$

and

$$\dot{\mathbf{p}}_i = -\frac{m_i\mathbf{q}_i}{q_i^3}. \qquad (i = 1, 2) \tag{42}$$

By (40) and (42), a first integral of these equations is

$$\mathbf{q}_1 + \mathbf{q}_1' = \mathbf{q}_2 + \mathbf{q}_2',$$

whence it follows that Equations (36) may be inverted and we can recover the old coordinate vector by such an equation as

$$\mathbf{q}' = \mathbf{q}_1 + \mathbf{q}_1'. \tag{43}$$

Then it is easy to establish that $\mathbf{q}'$ and $\mathbf{p}'$, defined by (43) and (38) respectively, satisfy Hamilton's equations for the Hamiltonian (35) with initial values $\mathbf{q}_0'$ and $\mathbf{p}_0'$, provided that $\mathbf{q}_i$ and $\mathbf{p}_i$ $(i=1, 2)$ are solutions of the system (40)–(42).

As before we denote by h the value of H on the solution path of interest, but it is no longer constant. In fact we deduce from (39) and (42) that

$$\dot{h} = -\ddot{\mathbf{q}}_1'^T\mathbf{p}_1 - \ddot{\mathbf{q}}_2'^T\mathbf{p}_2. \tag{44}$$

Now, in the centre of mass frame we may write $m_1\mathbf{q}_1' + m_2\mathbf{q}_2' = \mathbf{0}$, while if we set $\mathbf{q}_3 \equiv$ $\equiv \mathbf{q}_1' - \mathbf{q}_2'$ and $q_3 \equiv |\mathbf{q}_3|$ we will have

$$\ddot{\mathbf{q}}_3 = -\frac{(m_1 + m_2)\,\mathbf{q}_3}{q_3^3},$$

since the relative motion of the two primaries is Keplerian. Hence (44) may be written

$$\dot{h} = \frac{1}{q_3^3}\,\mathbf{q}_3^T(m_2\mathbf{p}_1 - m_1\mathbf{p}_2), \tag{45}$$

an equation which is identical with (30) apart from changes of notation, and which clarifies the meaning of k and the need for its introduction. The quantity h is not quite the energy per unit mass of the third body, but differs from this by the value of the quantity $\partial W/\partial t$.

The further progress of the derivation resembles so closely the corresponding work in Section 2 that only the results will be stated. We introduce a new time-variable by

$$\mathrm{d}t = R_1 R_2 R_3 \, \mathrm{d}\tau , \tag{46}$$

where $R_i \equiv q_i$ $(i=1, 2, 3)$ and perform KS-transformations on the coordinates. The Hamiltonian takes the form

$$\begin{aligned} \Gamma^* = &- hR_1R_2R_3 + \tfrac{1}{8}R_3(R_2\mathbf{P}_1^2 + R_1\mathbf{P}_2^2) + \tfrac{1}{16}R_3\mathbf{P}_1^T\mathbf{A}_1\mathbf{A}_2^T\mathbf{P}_2 - \\ &- R_3(m_1R_2 + m_2R_1) + \frac{1}{4}\frac{R_3}{m_1+m_2}\dot{\mathbf{q}}_3^T(-m_2R_2\mathbf{A}_1^T\mathbf{P}_1 + m_1R_1\mathbf{A}_2^T\mathbf{P}_2) \end{aligned}$$

and the equations of motion are

$$\mathbf{Q}_i' = \frac{\partial \Gamma^*}{\partial \mathbf{P}_i}, \quad \mathbf{P}_i' = -\frac{\partial \Gamma^*}{\partial \mathbf{Q}_i}, \qquad (i=1, 2)$$

a prime now denoting differentiation with respect to τ. These may be compared with the result of setting $m_3 = 0$ in Equations (27) and (32), the differences being attributable to changes of notation, and the use there of 'regularised' coordinates and momenta in place of $\mathbf{q}_3$ and $\dot{\mathbf{q}}_3$. Substituting for $\mathbf{p}_1$ and $\mathbf{p}_2$ in (45) by their expressions in terms of regularised variables, and using (46), we quickly obtain an equation for h' which is regular during collisions between the massless body and either primary. The equations of motion remain regular even if we omit the factor R_3 from (46) and from Γ^*.

For most investigations into the restricted problem it would be preferable to work in a system of coordinates in which the primaries are at rest. The principles used in the second half of the present section are easily applied to treat the corresponding Hamiltonian, but we shall not stop to give the details, for an alternative global regularisation of this problem already exists (Waldvogel, 1967).

As already mentioned, the extension of the present derivation to the case in which there is an arbitrary number of massive bodies is readily made. Such a formulation should be of value in the computation of orbits for a probe suffering successive close encounters with different planets, for example.

5. The Perturbed Three-Body Problem

Having discussed the special features required in the restricted problem, we now suppose that it is desired to regularise the equations for the relative motion of three particles with non-zero mass perturbed by $(N-3)$ others, where $N \geqslant 4$. In this case we start with the Hamiltonian

$$H' = \sum_{i=1}^{3} \frac{1}{2m_i} \mathbf{p}_i'^2 - \frac{m_2 m_3}{|\mathbf{q}_2' - \mathbf{q}_3'|} - \frac{m_3 m_1}{|\mathbf{q}_3' - \mathbf{q}_1'|} - \frac{m_1 m_2}{|\mathbf{q}_1' - \mathbf{q}_2'|} + \mathscr{R}' + \mathscr{T}_1' + \mathscr{R}_1', \tag{47}$$

where

$$\mathscr{R}' \equiv - \sum_{i=1}^{3} \sum_{j=4}^{N} \frac{m_i m_j}{|\mathbf{q}_i' - \mathbf{q}_j'|},$$

$$\mathscr{T}_1' \equiv \sum_{i=4}^{N} \frac{1}{2m_i} \mathbf{p}_i'^2, \tag{48}$$

$$\mathscr{R}_1' \equiv - \frac{1}{2} \sum_{i=4}^{N} \sum_{\substack{j=4 \\ j \neq i}}^{N} \frac{m_i m_j}{|\mathbf{q}_i' - \mathbf{q}_j'|},$$

and the notation is an obvious extension of that used in (1).

Following closely the development of section two, we introduce new coordinates

$$\mathbf{q}_1 = \mathbf{q}_2' - \mathbf{q}_3', \qquad (*) \tag{49}$$

$$\langle \mathbf{q} \rangle = \frac{1}{M} \sum_{i=1}^{3} m_i \mathbf{q}_i', \tag{50}$$

$$\mathbf{q}_i = \mathbf{q}_i', \qquad (i = 4, \ldots, N) \tag{51}$$

and new momenta satisfying

$$\mathbf{p}_1' = - \mathbf{p}_2 + \mathbf{p}_3 + \frac{m_1}{M} \langle \mathbf{p} \rangle \qquad (*) \tag{52}$$

and

$$\mathbf{p}_i' = \mathbf{p}_i. \qquad (i = 4, \ldots, N) \tag{53}$$

By comparison with (10), we see that (49) and (50) may be inverted to give

$$\mathbf{q}_1' = \frac{1}{M} (m_2 \mathbf{q}_3 - m_3 \mathbf{q}_2) + \langle \mathbf{q} \rangle, \qquad (*) \tag{54}$$

a result which is needed in the right-hand side of (48). The new Hamiltonian is found by substitution of (49) and (51)–(54) into (47), the result being

$$H = \frac{1}{2\mu_{23}} \mathbf{p}_1^2 + \frac{1}{2\mu_{31}} \mathbf{p}_2^2 + \frac{1}{2\mu_{12}} \mathbf{p}_3^2 + \frac{1}{2M} \langle \mathbf{p} \rangle^2 -$$
$$- \left(\frac{1}{m_1} \mathbf{p}_2^T \mathbf{p}_3 + \frac{1}{m_2} \mathbf{p}_3^T \mathbf{p}_1 + \frac{1}{m_3} \mathbf{p}_1^T \mathbf{p}_2 \right) -$$
$$- \frac{m_2 m_3}{q_1} - \frac{m_3 m_1}{q_2} - \frac{m_1 m_2}{q_3} + \mathscr{R} + \mathscr{T}_1 + \mathscr{R}_1,$$

where

$$\mathscr{R} \equiv -\sum_{j=4}^{N} m_j \left\{ \frac{m_1}{\left|\frac{m_2}{M}\mathbf{q}_3 - \frac{m_3}{M}\mathbf{q}_2 + \langle \mathbf{q} \rangle - \mathbf{q}_j\right|} + \right.$$
$$\left. + \frac{m_2}{\left|\frac{m_3}{M}\mathbf{q}_1 - \frac{m_1}{M}\mathbf{q}_3 + \langle \mathbf{q} \rangle - \mathbf{q}_j\right|} + \frac{m_3}{\left|\frac{m_1}{M}\mathbf{q}_2 - \frac{m_2}{M}\mathbf{q}_1 + \langle \mathbf{q} \rangle - \mathbf{q}_j\right|} \right\}, \tag{55}$$

$$\mathscr{T}_1 \equiv \sum_{i=4}^{N} \frac{1}{2m_i}\mathbf{p}_i^2, \tag{56}$$

and

$$\mathscr{R}_1 \equiv -\frac{1}{2}\sum_{i=4}^{N}\sum_{\substack{j=4 \\ j\neq i}}^{N} \frac{m_i m_j}{|\mathbf{q}_i - \mathbf{q}_j|}. \tag{57}$$

We may again show directly that, if we solve Hamilton's equations with this Hamiltonian and with appropriate initial conditions, then we may recover solutions to the perturbed three-body problem specified by the Hamiltonian (47). Since the formulae are a little more complicated than in Section 2, however, we prefer now to appeal to the much more general result proved in the Appendix. Equations governing $\mathbf{q}_i$, $\mathbf{p}_i$ $(i=4\ldots, N)$, $\langle \mathbf{q} \rangle$ and $\langle \mathbf{p} \rangle$ are most conveniently obtained at this stage, since we have

$$\dot{\mathbf{q}}_i = \frac{1}{m_i}\mathbf{p}_i, \qquad \dot{\mathbf{p}}_i = -\frac{\partial}{\partial \mathbf{q}_i}(\mathscr{R} + \mathscr{R}_1) \qquad (i = 4, \ldots, N) \tag{58}$$

and

$$\langle \dot{\mathbf{q}} \rangle = \frac{1}{M}\langle \mathbf{p} \rangle, \qquad \langle \dot{\mathbf{p}} \rangle = -\frac{\partial \mathscr{R}}{\partial \langle \mathbf{q} \rangle}.$$

The last two equations yield the perturbing effect of the $(N-3)$ bodies on the centre of mass of the first three, and it is convenient here to obtain also their effect on the 'internal' energy of the triple system, denoted by h_3 and defined by

$$h_3 \equiv \frac{1}{2\mu_{23}}\mathbf{p}_1^2 + \frac{1}{2\mu_{31}}\mathbf{p}_2^2 + \frac{1}{2\mu_{12}}\mathbf{p}_3^2 - \left(\frac{1}{m_1}\mathbf{p}_2^T\mathbf{p}_3 + \frac{1}{m_2}\mathbf{p}_3^T\mathbf{p}_1 + \frac{1}{m_3}\mathbf{p}_1^T\mathbf{p}_2\right) -$$
$$- \frac{m_2 m_3}{q_1} - \frac{m_3 m_1}{q_2} - \frac{m_1 m_2}{q_3}.$$

For this purpose only we need the Hamiltonian equations for these three particles, and they are

$$\dot{\mathbf{q}}_1 = \frac{1}{\mu_{23}}\mathbf{p}_1 - \frac{1}{m_2}\mathbf{p}_3 - \frac{1}{m_3}\mathbf{p}_2 \qquad (*)$$

and

$$\dot{\mathbf{p}}_1 = -\frac{m_2 m_3}{q_1^3}\,\mathbf{q}_1 - \frac{\partial \mathscr{R}}{\partial \mathbf{q}_1}. \qquad (*)$$

After a little reduction, we arrive at the result

$$h_3 = -\dot{\mathbf{q}}_1^T \frac{\partial \mathscr{R}}{\partial \mathbf{q}_1} - \dot{\mathbf{q}}_2^T \frac{\partial \mathscr{R}}{\partial \mathbf{q}_2} - \dot{\mathbf{q}}_3^T \frac{\partial \mathscr{R}}{\partial \mathbf{q}_3}. \qquad (59)$$

We now apply KS-transformations to the vectors $\mathbf{q}_i (i=1, 2, 3)$, just as in Section 2, and obtain a Hamiltonian, $\mathscr{H}$, which is given by

$$\mathscr{H} = \mathscr{H}_3 + \frac{1}{2M}\langle \mathbf{p} \rangle^2 + \mathscr{R} + \mathscr{T}_1 + \mathscr{R}_1,$$

where

$$\begin{aligned}\mathscr{H}_3 \equiv & \frac{1}{8\mu_{23}R_1}\mathbf{P}_1^T\mathbf{P}_1 + \frac{1}{8\mu_{31}R_2}\mathbf{P}_2^T\mathbf{P}_2 + \frac{1}{8\mu_{12}R_3}\mathbf{P}_3^T\mathbf{P}_3 - \\ & - \frac{1}{16}\left(\frac{1}{m_1R_2R_3}\mathbf{P}_2^T\mathbf{A}_2\mathbf{A}_3^T\mathbf{P}_3 + \frac{1}{m_2R_3R_1}\mathbf{P}_3^T\mathbf{A}_3\mathbf{A}_1^T\mathbf{P}_1 + \right. \\ & \left. + \frac{1}{m_3R_1R_2}\mathbf{P}_1^T\mathbf{A}_1\mathbf{A}_2^T\mathbf{P}_2\right) - \frac{m_2m_3}{R_1} - \frac{m_3m_1}{R_2} - \frac{m_1m_2}{R_3},\end{aligned}$$

and $\mathscr{R}$, $\mathscr{T}_1$ and $\mathscr{R}_1$ are given by (55)–(57). If we wish, however, (55) may be expressed in terms of $\mathbf{Q}_i$ rather than $\mathbf{q}_i$ $(i=1, 2, 3)$ by making use of (17). $\mathscr{H}_3$ has the value h_3 on the solution path.

As before, we denote by h the value of $\mathscr{H}$ on the solution path of interest. If we perform the same transformation of the independent variable as the first suggested previously, i.e. (20), the appropriate Hamiltonian is

$$\Gamma^* = \left(\mathscr{H}_3 + \frac{1}{2M}\langle \mathbf{p} \rangle^2 + \mathscr{R} + \mathscr{T}_1 + \mathscr{R}_1 - h\right) R_1R_2R_3.$$

Hamilton's equations are now of the form

$$\mathbf{Q}_i' = \frac{\partial}{\partial \mathbf{P}_i}(R_1R_2R_3\mathscr{H}_3) \qquad (i=1, 2, 3)$$

and

$$\begin{aligned}\mathbf{P}_i' = & -\frac{\partial}{\partial \mathbf{Q}_i}(\mathscr{H}_3R_1R_2R_3) - R_1R_2R_3\frac{\partial \mathscr{R}}{\partial \mathbf{Q}_i} - \\ & - \left(\frac{1}{2M}\langle \mathbf{p} \rangle^2 + \mathscr{R} + \mathscr{T}_1 + \mathscr{R}_1 - h\right)\frac{\partial}{\partial \mathbf{Q}_i}(R_1R_2R_3). \qquad (i=1, 2, 3)\end{aligned} \qquad (60)$$

The middle term on the right-hand side may be written as

$$-R_1R_2R_3\mathbf{A}_i\frac{\partial \mathscr{R}}{\partial \mathbf{q}_i}$$

if desired, by (17) and (18). Also, as it stands at present, computation of the last term of (60) involves a number of operations of order N^2, because of the presence of $\mathscr{R}_1$.

Since this is time-consuming, it is profitable to note that the coefficient of this term is merely $-h_3$, which can be obtained by integration of (59). In turn, this equation can be written in the regular and more useful form

$$h_3' = -\mathbf{Q}_1'^T \mathbf{A}_1 \frac{\partial \mathscr{R}}{\partial \mathbf{q}_1} - \mathbf{Q}_2'^T \mathbf{A}_2 \frac{\partial \mathscr{R}}{\partial \mathbf{q}_2} - \mathbf{Q}_3'^T \mathbf{A}_3 \frac{\partial \mathscr{R}}{\partial \mathbf{q}_3},$$

using (17) and (18). Appropriate modifications to some of these formulae would be required if some other time-transformation, such as (22), were adopted.

If N is large, most of the time expended in a practical application would be devoted to computation of the perturbing forces, i.e. the terms $\partial \mathscr{R}/\partial \mathbf{q}_i$ $(i=1, 2, 3)$, whether the present method or that of Aarseth and Zare was used. In comparing the numerical results of Section 3, it is therefore justifiable to take the number of integration steps as a realistic measure of the computational effort in each integration. This consideration reduces the slight but significant relative inefficiency, for certain problems, of the method discussed in this paper, to which attention was drawn during discussion of the Pythagorean and 'realistic' problems.

6. Global Regularisation of the N-Body Problem

So far we have generalised the regularisation of Section 2 in one direction, having shown how it may be modified to cope with perturbed motion of three bodies. Here we generalise in another direction, showing how the same principles may be used to reach another desirable goal – a global regularisation of the N-body problem, where $N \geqslant 2$. We shall be able to see that, when $N=2$, the new method reduces to KS-regularisation, and when $N=3$, it reduces to the technique developed in Section 2 of this paper. Our basic Hamiltonian is now

$$H' = \sum_{i=1}^{N} \frac{1}{2m_i} \mathbf{p}_i'^2 - \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{m_i m_j}{|\mathbf{q}_i' - \mathbf{q}_j'|}, \tag{61}$$

the notation being the necessary extension of that used in (1).

We introduce new variables with the definitions

$$\mathbf{q}_{ij} \equiv \mathbf{q}_i' - \mathbf{q}_j' \qquad (1 \leqslant i < j \leqslant N)$$

and

$$\langle \mathbf{q} \rangle \equiv \frac{1}{M} \sum_{i=1}^{N} m_i \mathbf{q}_i', \tag{62}$$

where now $M \equiv \sum_{i=1}^{N} m_i$. The conjugate momenta are defined to satisfy the following generalisation of (5) or, if preferred, the Equation (A5) of the general result in the Appendix to this paper:

$$\mathbf{p}_i' = \sum_{j=i+1}^{N} \mathbf{p}_{ij} - \sum_{j=1}^{i-1} \mathbf{p}_{ji} + \frac{m_i}{M}\langle \mathbf{p} \rangle, \qquad (i = 1, \ldots, N) \tag{63}$$

it being understood that an empty sum is taken to have the value zero. If these equations are substituted into (61) there results, after a little manipulation, the new Hamiltonian

$$H = \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{1}{2\mu_{ij}} \mathbf{p}_{ij}^2 + \sum_{i=1}^{N} \frac{1}{m_i} \Bigg(\sum_{j=i+1}^{N-1} \sum_{k=j+1}^{N} \mathbf{p}_{ij}^T \mathbf{p}_{ik} + \sum_{j=1}^{i-2} \sum_{k=j+1}^{i-1} \mathbf{p}_{ji}^T \mathbf{p}_{ki} - \sum_{j=i+1}^{N} \sum_{k=1}^{i-1} \mathbf{p}_{ij}^T \mathbf{p}_{ki} \Bigg) + \frac{1}{2M} \langle \mathbf{p} \rangle^2 - \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{m_i m_j}{q_{ij}},$$

where $\mu_{ij} \equiv m_i m_j/(m_i + m_j)$ and $q_{ij} \equiv |\mathbf{q}_{ij}|$.

As before, we may drop the term in $\langle \mathbf{p} \rangle$ if we are content to refer the particles to the frame in which their centre of mass is at rest at the origin, and this we do. Again, solution of Hamilton's equations with this Hamiltonian and appropriate initial conditions yields solutions of the N-body problem by means of (63) and a suitable inversion of (62). In fact we easily find that

$$\sum_{j=i+1}^{N} m_j \mathbf{q}_{ij} = \mathbf{q}_i' \sum_{j=i+1}^{N} m_j - \sum_{j=i+1}^{N} m_j \mathbf{q}_j'$$

and

$$\sum_{j=1}^{i-1} m_j \mathbf{q}_{ji} = \sum_{j=1}^{i-1} m_j \mathbf{q}_j' - \mathbf{q}_i' \sum_{j=1}^{i-1} m_j,$$

whence

$$\mathbf{q}_i' = \langle \mathbf{q} \rangle + \frac{1}{M} \Bigg(\sum_{j=i+1}^{N} m_j \mathbf{q}_{ij} - \sum_{j=1}^{i-1} m_j \mathbf{q}_{ji} \Bigg), \qquad (i = 1, \ldots, N)$$

although we normally take $\langle \mathbf{q} \rangle = \mathbf{0}$.

Following the course laid out for the three-body problem in Section 2, we perform a KS-transformation on each of the new coordinate vectors, obtaining a set of 4-vectors $\mathbf{Q}_{ij}$ and their conjugate momenta $\mathbf{P}_{ij}$ $(1 \leqslant i < j \leqslant N)$. The Hamiltonian consequently assumes the form

$$H = \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{1}{8\mu_{ij} R_{ij}} \mathbf{P}_{ij}^T \mathbf{P}_{ij} + \sum_{i=1}^{N} \frac{1}{16 m_i} \Bigg(\sum_{j=i+1}^{N-1} \sum_{k=j+1}^{N} \frac{1}{R_{ij} R_{ik}} \times$$
$$\times \mathbf{P}_{ij}^T \mathbf{A}_{ij} \mathbf{A}_{ik}^T \mathbf{P}_{ik} + \sum_{j=1}^{i-2} \sum_{k=j+1}^{i-1} \frac{1}{R_{ji} R_{ki}} \mathbf{P}_{ji}^T \mathbf{A}_{ji} \mathbf{A}_{ki}^T \mathbf{P}_{ki} - \sum_{j=i+1}^{N} \sum_{k=1}^{i-1} \frac{1}{R_{ij} R_{ki}} \times$$
$$\times \mathbf{P}_{ij}^T \mathbf{A}_{ij} \mathbf{A}_{ki}^T \mathbf{P}_{ki} \Bigg) - \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{m_i m_j}{R_{ij}},$$

where $R_{ij} \equiv q_{ij}$, and the set of matrices A_{ij} is defined in a manner analogous to (18).

Having completed the transformation of the dependent variables, we adopt the obvious generalisation of (20), namely

$$d\tau = \frac{dt}{\prod_{i=1}^{N-1} \prod_{j=i+1}^{N} R_{ij}}, \tag{64}$$

although others might be contemplated. Then, if h is the value of $\mathscr{H}$ on some solution path, we have a new Hamiltonian defined by

$$\Gamma^* \equiv (\mathscr{H} - h) \prod_{i=1}^{N-1} \prod_{j=i+1}^{N} R_{ij}.$$

Hamilton's equations, now in the form

$$\mathbf{Q}'_{ij} = \frac{\partial \Gamma^*}{\partial \mathbf{P}_{ij}}, \qquad \mathbf{P}'_{ij} = -\frac{\partial \Gamma^*}{\partial \mathbf{Q}_{ij}}, \quad (1 \leqslant i < j \leqslant N) \tag{65}$$

are regular with respect to collisions between any pair of the N bodies, provided that two or more collisions do not occur simultaneously.

In practical terms these equations rapidly become inefficient as N increases. There are $\frac{1}{2}N(N-1)$ pairs of values of the subscripts i and j and for each such pair, (65) is a set of eight first-order differential equations. With (64), we have $4N(N-1)+1$ equations in all. By contrast, the original Hamiltonian equations derived from (61) form a set of $6N$ first-order equations, or $6(N-1)$ if the positions and velocities are referred to the centre of mass. With $N=2$, for which the present regularisation reduces to that of Kustaanheimo and Stiefel, regularisation increases the number of equations to be integrated from 6 to 9, and for $N=3$, i.e. the regularisation of Section 2, from 12 to 25. It is also important to note that the number of terms on the right-hand side of the second member of (65) grows more rapidly with increasing N than the number of terms in the usual Newtonian equations.

The case in which one of the bodies is massless is a straightforward generalisation of the material in the second half of Section 4, as mentioned there. Further, there would seem to be no difficulty in extending the discussions of Section 5 and the present section to deal with the regularisation of the relative motion of n bodies perturbed by $(N-n)$ others, where $N > n \geqslant 2$. Indeed, it should be possible to perform simultaneous regularisation on any number of subsystems of any size in any N-body system, but whether the efficiency of a computer programme developed along such lines would justify the effort involved is, perhaps, another matter.

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Appendix: Enlargement of a Dynamical System

Consider a dynamical system with generalised coordinates q'_i $(i=1,\ldots,n')$ and conjugate momenta p'_i $(i=1,\ldots,n')$, the Hamiltonian being $H'(q',p')$. Note that variables with missing subscripts are used to denote the corresponding vector; thus, $q' \equiv \{q'_i\}$.

We suppose that there exist functions q_i $(i=1,\ldots,n,\ \geqslant n')$ of n' variables such that

(i) the dependence of H' on q' is expressible in terms of the functions q, i.e.

$$H'(q',p') \equiv H''(q(q'),p'), \tag{A1}$$

where H'' is some function, and

(ii) the rank of the matrix $\{\partial q_i/\partial q'_j\}$ is maximal, i.e. equal to n'.

As an example, consider the Hamiltonian (1) corresponding to the three-body problem and, as the functions q, those defined by (4). We would then have $n'=9$, $n=12$, and

$$H''(q,p') \equiv \sum_{i=1}^{3} \frac{1}{2m_i}\mathbf{p}_i'^2 - \frac{m_2m_3}{q_1} - \frac{m_3m_1}{q_2} - \frac{m_1m_2}{q_3},$$

where $q_i \equiv |\mathbf{q}_i|$ for $i=1, 2, 3$. The matrix $\{\partial q_i/\partial q'_j\}$ is

$$\begin{pmatrix} \mathbf{0} & \mathbf{I} & -\mathbf{I} \\ -\mathbf{I} & \mathbf{0} & \mathbf{I} \\ \mathbf{I} & -\mathbf{I} & \mathbf{0} \\ \frac{m_1}{M}\mathbf{I} & \frac{m_2}{M}\mathbf{I} & \frac{m_3}{M}\mathbf{I} \end{pmatrix},$$

where $\mathbf{I}$ and $\mathbf{0}$ are, respectively, the 3×3 unit and zero matrices, and its rank is 9.

By the second condition above, there is a subset q_{i_j} $(j=1,\ldots,n';\ 1\leqslant i_1<\cdots<i_j<i_{j+1}<\cdots<i_{n'}\leqslant n)$ of the functions q_i such that the Jacobian $|\partial q_{i_j}/\partial q'_k|$ is non-zero. Invoking the inverse function theorem, we establish that there exists a set of n' functions q_i^* $(i=1,\ldots,n')$ of n' variables with the property that

$$q_k^*(q_{i_j}(q')) \equiv q'_k, \tag{A2}$$

whence

$$\sum_{j=1}^{n'} \left\{\frac{\partial}{\partial q_{i_j}} q_k^*(q_{i_j}(q'))\right\} \frac{\partial}{\partial q'_l}\, q_{i_j}(q') = \delta_{kl}.$$

We can now regard the functions q^* as functions of n variables, although in fact independent of $(n-n')$ of them, and these conditions become

$$q_k^*(q(q')) \equiv q_k' \tag{A3}$$

and

$$\sum_{j=1}^{n} \left\{ \frac{\partial}{\partial q_j} q_k^*(q(q')) \right\} \frac{\partial}{\partial q_l'} q_j(q') = \delta_{kl}. \tag{A4}$$

Supported by this informal discussion on existence, we now let q_i^* $(i=1, \ldots, n')$ be *any* set of functions of n variables satisfying (A3) and (A4). For the three-body problem, the functions defined by (10) enjoy these properties.

By analogy with the theory of point transformations (e.g. Goldstein, 1950, p. 244), which is the case $n=n'$, we introduce n new variables $p_i (i=1, \ldots, n)$ satisfying

$$p_i' = \sum_{j=1}^{n} \frac{\partial q_j}{\partial q_i'} p_j. \qquad (i = 1, \ldots, n') \tag{A5}$$

These are n' equations in n unknowns and, by condition (ii) above, such functions can be found. In fact our choice is generally very wide. We also introduce a new Hamiltonian defined by

$$H(q, p) \equiv H''\left(q, \sum_{j=1}^{n} p_j \frac{\partial}{\partial q'} q_j(q^*(q))\right). \tag{A6}$$

It will be noted that the derivatives $\partial q_j / \partial q'$ are evaluated at $q_i' \equiv q_i^*(q)$, in order that the Hamiltonian be dependent only on the new variables. Equations (A5) and (A6) correspond, in the situation treated in Section 2, to (5) and (6).

We now suppose that q_i' $(i=1, \ldots, n')$ are functions of an independent variable, t, the time, such that

$$\begin{aligned} \dot{q}_i' &= \frac{\partial}{\partial p_i'} H'(q', p') \\ \dot{p}_i' &= -\frac{\partial}{\partial q_i'} H'(q', p'), \end{aligned} \qquad (i = 1, \ldots, n') \tag{A7}$$

where a dot denotes differentiation with respect to t. Hence, by (A1),

$$\dot{q}_i' = \frac{\partial}{\partial p_i'} H''(q(q'), p'). \qquad (i = 1, \ldots, n') \tag{A8}$$

Our next step is to obtain an equation satisfied by q provided that (A5) holds. In fact, by (A8), we have

$$\dot{q}_i = \sum_{j=1}^{n'} \frac{\partial q_i}{\partial q'_j} \frac{\partial}{\partial p'_j} H''(q, p'),$$

$$= \frac{\partial}{\partial p_i} H(q, p), \qquad (i = 1, \ldots, n)$$

by (A6).

We have now to ensure that (A5) holds for all time t. Since these equations are not sufficient to determine the variables p, it is not possible to obtain any differential equation which the new momenta necessarily satisfy. Instead, we write down an equation which is sufficient to ensure that (A5) is obeyed. In fact, let us suppose that p_i $(i=1, \ldots, n)$ are functions of t satisfying

$$\dot{p}_i = -\frac{\partial}{\partial q_i} H(q, p) \qquad (i=1, \ldots, n) \tag{A9}$$

and set

$$p_j^* \equiv \sum_{i=1}^{n} \left\{ \frac{\partial}{\partial q'_j} q_i(q^*(q)) \right\} p_i. \qquad (j = 1, \ldots, n') \tag{A10}$$

We also suppose that $p_j^* = p'_j$ initially. Then we see that

$$\dot{p}_j^* = \sum_{i=1}^{n} \sum_{k=1}^{n'} \frac{\partial^2 q_i}{\partial q'_k \partial q'_j} p_i \frac{\partial}{\partial p'_k} H''(q, p') - \sum_{i=1}^{n} \frac{\partial q_i}{\partial q'_j} \frac{\partial}{\partial q_i} H(q, p), \tag{A11}$$

by (A8) and (A9). The second term here is

$$-\sum_{i=1}^{n} \frac{\partial q_i}{\partial q'_j} \left\{ \frac{\partial}{\partial q_i} H''(q, p^*) + \sum_{k=1}^{n'} \frac{\partial p_k^*}{\partial q_i} \frac{\partial}{\partial p'_k} H''(q, p^*) \right\}$$

by (A6), and this is equal to

$$-\frac{\partial}{\partial q'_j} H'(q, p^*) - \sum_{i=1}^{n} \sum_{k=1}^{n'} \frac{\partial q_i}{\partial q'_j} \left\{ \frac{\partial}{\partial p'_k} H''(q, p^*) \right\} \sum_{l=1}^{n} \sum_{m=1}^{n'} \frac{\partial^2 q_l}{\partial q'_m \partial q'_k} p_l \frac{\partial q_m^*}{\partial q_i},$$

by (A1) and (A10). Using (A4), substituting back into (A11) and using (A7), we observe that

$$\dot{p}_j^* - \dot{p}'_j = -\frac{\partial}{\partial q'_j} H'(q', p^*) + \frac{\partial}{\partial q'_j} H'(q', p') +$$

$$+ \sum_{i=1}^{n} \sum_{k=1}^{n'} \frac{\partial^2 q_i}{\partial q'_k \partial q'_j} p_i \left\{ \frac{\partial}{\partial p'_k} H''(q, p') - \frac{\partial}{\partial p'_k} H''(q, p^*) \right\}.$$

Clearly, $p_j^* = p'_j$ is a solution of this equation and the initial conditions and, being an initial value problem, its solution is unique.

The results can be summarised as follows. We have shown that there exist functions $q_i, p_i (i=1, \ldots, n)$ of t satisfying appropriate initial conditions and the equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \qquad (i = 1, \ldots, n) \tag{A12}$$

and related to solutions of the system (A7) by Equations (A3) and (A5). The solution to this initial value problem is unique, and therefore solution of these equations allows us to recover p' and q' by (A3) and (A5). It must not be assumed that *every* solution of (A12) yields a solution to (A7) in this way, for we require that $q_i = q_i(q')$ $(i=1, \ldots, n)$ initially, for some given values q'_j $(j=1, \ldots, n')$, and that the initial values of p_i $(i=1, \ldots$ $\ldots, n)$ satisfy (A5) for given values of $p'_j (j=1, \ldots, n')$.

It will have been noticed that the structure of the proof just given differs significantly from that given for the special case in Section 2. The reason is that, in the case of the three-body problem, it was possible to write down an integral of the new Hamiltonian, viz. (9), and this step is difficult to generalise completely.

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